

*Revisiting the Role of Financial
Intermediaries' Leverage in Determining Their
Market Valuation*

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The Glory of Intermediary Pricing

- **Broker-Dealers as Marginal Investors**

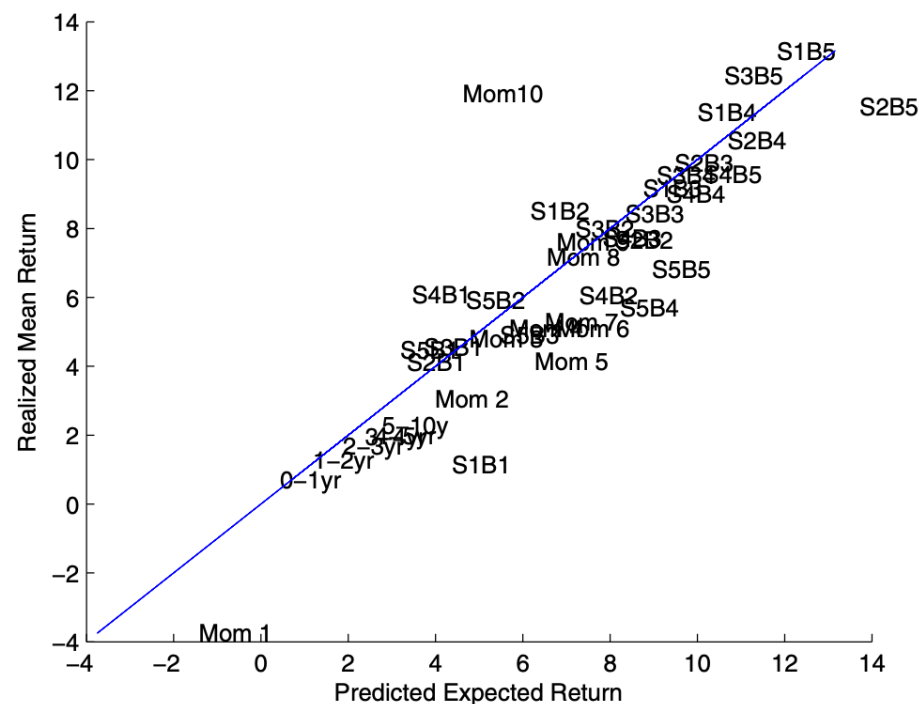
Intermediaries actively manage their balance sheets, making them key marginal pricers of assets.

- **Procyclical Leverage Mechanism**

Driven by market-based Value-at-Risk (VaR) constraints, leverage expands during booms and shrinks during busts.

- **Empirical Success (Pre-2008)**

Adrian, Etula, and Muir (2014) demonstrate that a single “Leverage Factor” perfectly captures the cross-sectional variation in expected returns ($R^2 \approx 77\%$).



- 25 Portfolios formed on Size and Book-to-Market
- 10 Portfolios formed on Momentum
- 6 Treasury Bond Portfolios

The Post-GFC Puzzle

- **The Anomaly**

In the modern era (post-2010), the cross-sectional pricing power of the leverage factor completely vanishes.

- **Statistical Collapse**

The estimated risk premium drops significantly, or even turns negative.

Adj. R^2 : 0.78 $\rightarrow$ 0.07

- **The Key Question**

Did regulation — not theory — break the leverage factor?

Pre- vs. Post-GFC

	Pre-GFC	Post-GFC
Sign of β	+	—
Adj. R^2	0.78	0.07
MAPE (%)	1.19	16.09

We: Regulation Broke the Signal, Not the Theory

- Theory is valid — but the system changed
- Pre-2010: market-driven VaR constraints
- Post-2010: rigid regulatory constraints (Basel III, SLR, Dodd-Frank)
- Factor failure = regulatory success

Pre-2010 Market-driven era

- ▶ **Constraint:** Market-based VaR limits
- ↓ **Mechanism:** Procyclical leverage expansion
(boom → vol.↓ → assets↑ → leverage↑)
- ↓ **Implication:** $LevFac_t \approx M_t$
(proxies marginal utility of wealth)

✓ **Good SDF proxy** Adj. $R^2 = 0.71$

Post-2010 Regulatory era

- ▶ **Constraint:** Basel III SLR / Dodd-Frank
(state-independent, binding)
- ↓ **Mechanism:** Leverage volatility suppressed
(~90% reduction in shock variance)
- ↓ **Implication:** $LevFac_t \not\approx M_t$
(leverage is noise, not signal)

× **Broken SDF proxy** Adj. $R^2 = 0.05$

Data

- **Vintage Data**

We reconstruct historical Flow of Funds (Z.1) series to accurately reflect real-time information, avoiding look-ahead bias from the Fed's ex-post revisions.

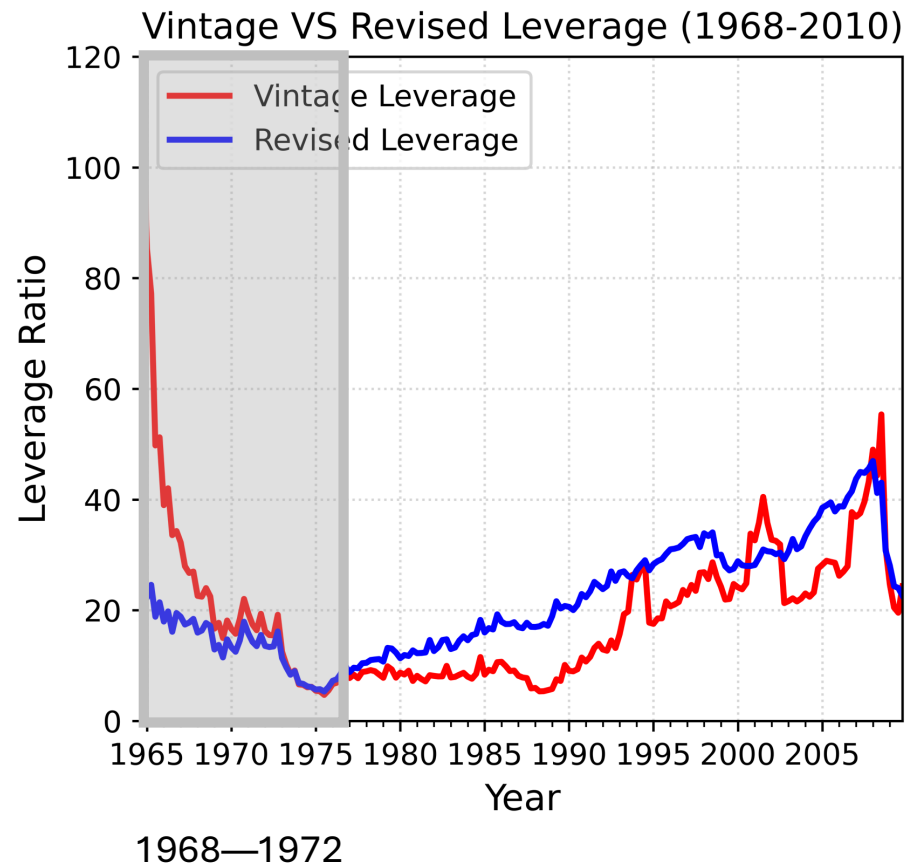
- **Revised Data**

Fed revises Z.1 ex-post — incorporates information unavailable at the time AEM (2014) uses vintage

Vintage leverage is more volatile — especially around 1968— 1972. We show results are sensitive to data revision.

$$Leverage_t = \frac{TotalAssets_t}{TotalAssets_t - TotalLiabilities_t}$$

$$LevFac_t = \Delta \ln(Leverage_t)^{SA}$$



Methodology

- **Standard two-pass cross-sectional regression**
 - Estimating unconditional, constant betas for each test asset
 - Estimating the market price of risk

$$R_{i,t}^e = \alpha_i + \beta_i' f_t + \epsilon_{i,t}, i = 1, \dots, N, t = 1, \dots, T$$

$$\overline{R_i^e} = \lambda_0 + \widehat{\beta_i'} \lambda + \alpha_i, i = 1, \dots, N$$

- **Test Assets (41 Portfolios)**

25 size and book-to-market portfolios,
10 momentum portfolios
6 Duration-sorted U.S. Treasury Bond Portfolios

- **Sample Split**

Pre-GFC: 1968–2009

Post-GFC Regulatory Era: 2010–2025

t : quarterly data

Finding 1a — Replication with Vintage Data

- **Replication succeeds with vintage data**
Using original vintage data, we successfully replicate the strong pre-crisis pricing performance.
- Vintage data: Adj. $R^2 = 0.78$
- 25 Portfolios formed on Size and Book-to-Market
- 10 Portfolios formed on Momentum
- 6 Treasury Bond Portfolios

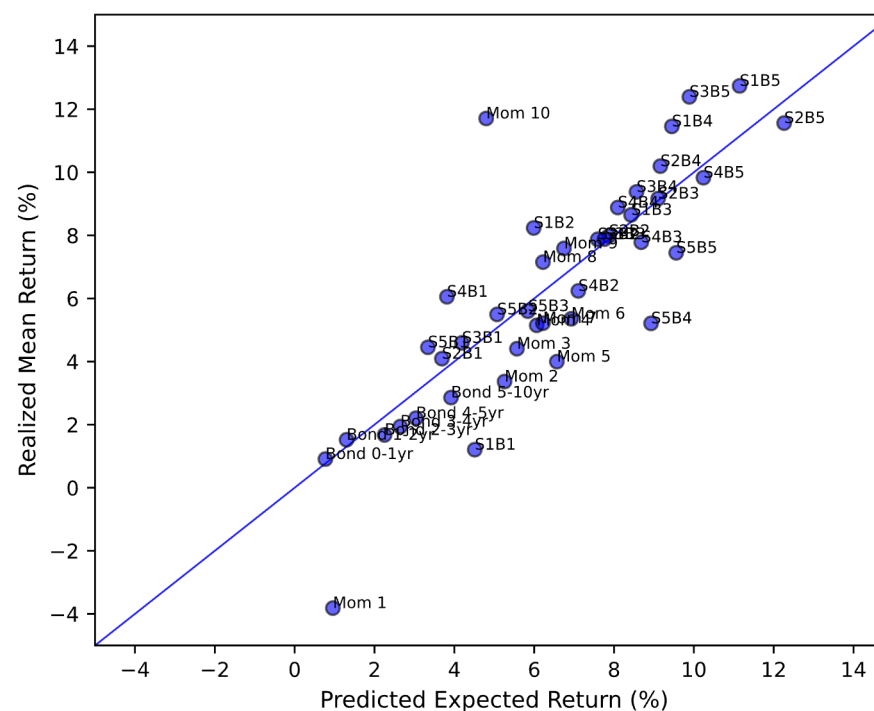


Figure A1. Realized versus predicted mean returns: leverage factor on Vintage Data (1968-2009).

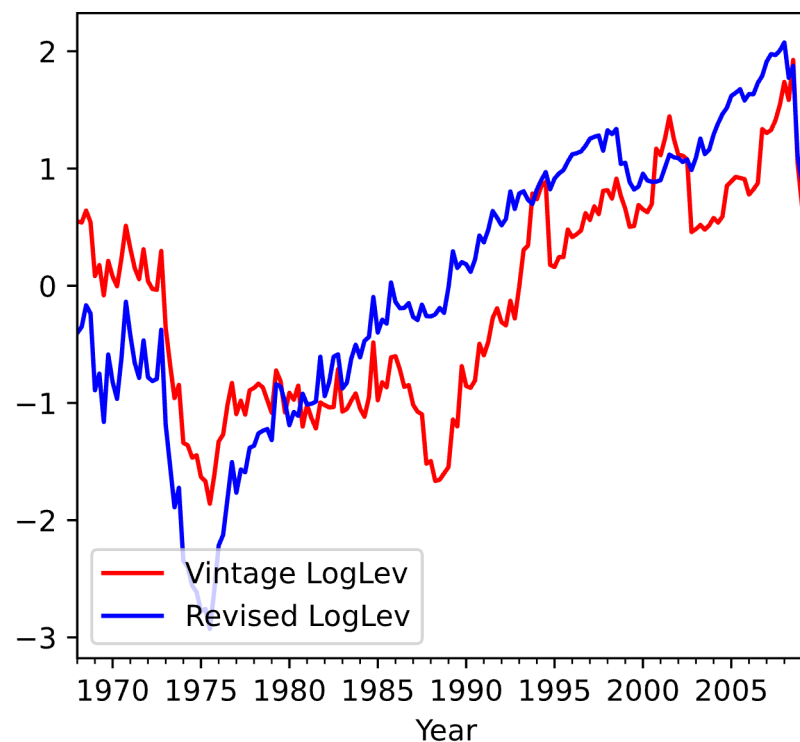
Finding 1b — Sensitivity to Data Revision

- **The Revision Effect**

Factor performance is highly sensitive to the Federal Reserve's retroactive adjustments.

- **Implication**

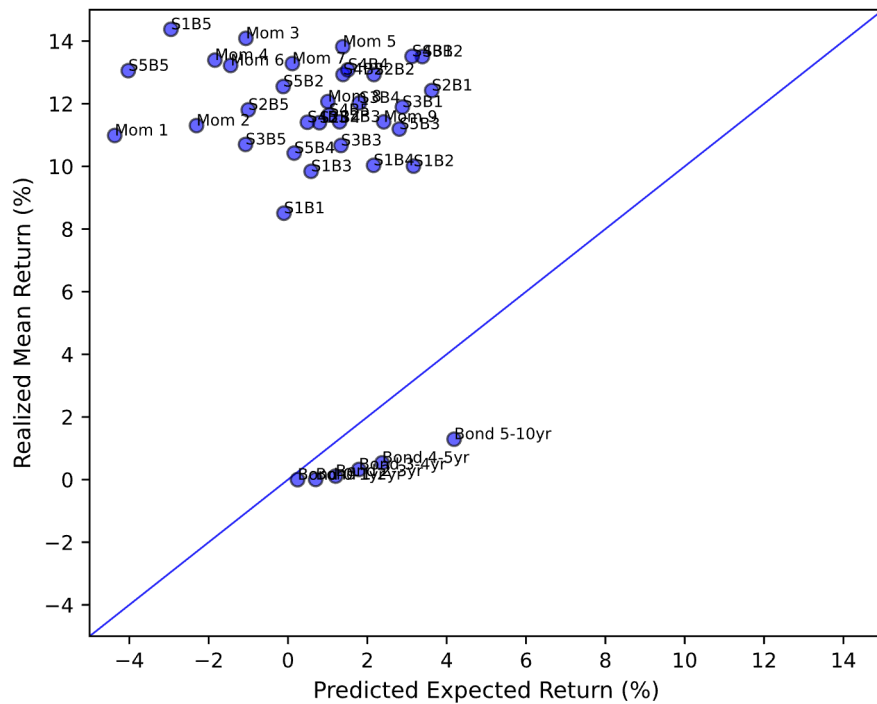
Evaluating the leverage factor with currently downloaded, heavily revised macroeconomic data significantly weakens its historical explanatory power.



Finding 2: LevFac Collapses, Benchmarks Hold

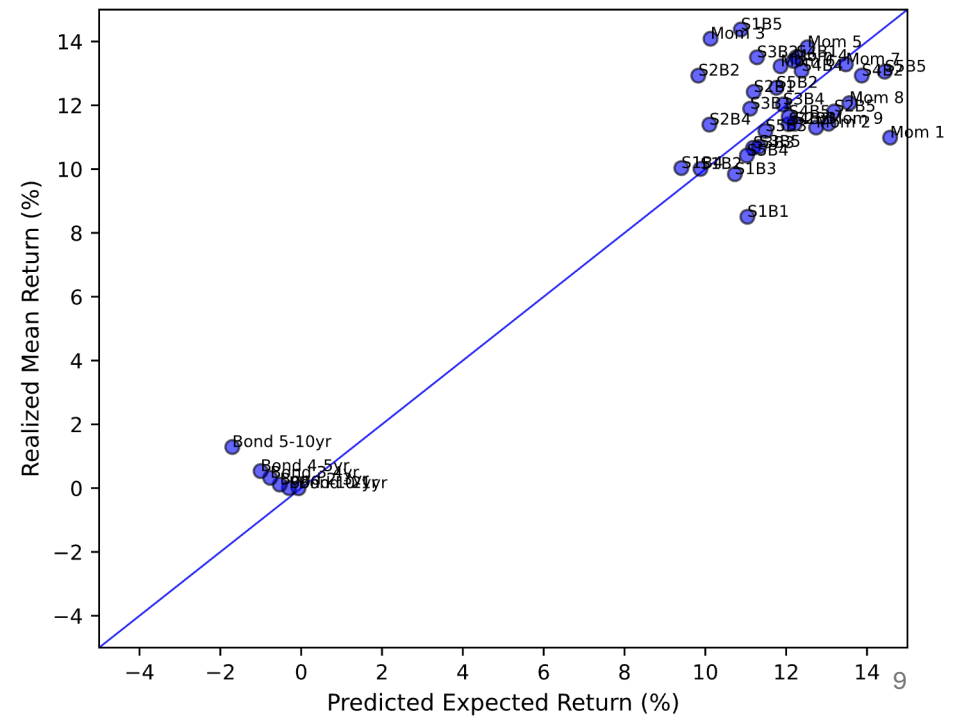
- **LevFac Fails in the Post-GFC Era**

Adj. R^2 collapses from **0.71** to **0.05**.



- **The Post-GFC Benchmark Comparison**

standard benchmarks (Fama-French 3-factor, Momentum, Bond PC1)



Cross-Asset Nuance: Bond-Equity Asymmetry

- **An Uneven Breakdown**

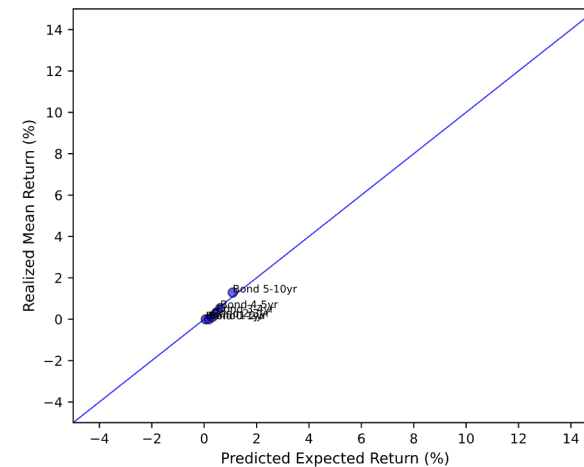
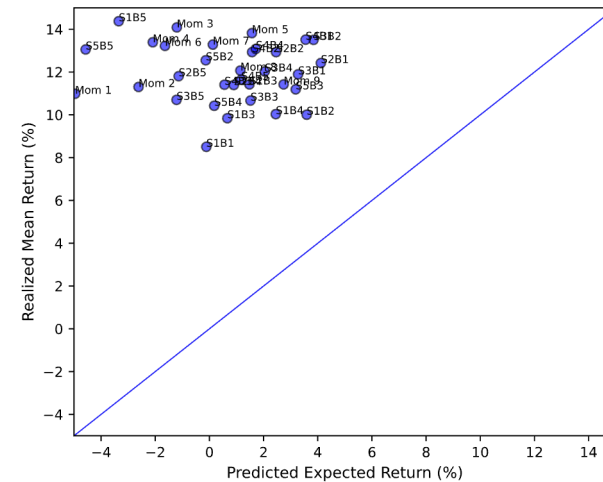
The post-crisis collapse is not uniform across different asset classes.

- **Equities (Complete Failure)**

Basel III penalizes active risk-taking → link severed

- **Treasury Bonds (Survival)**

Duration risk unaffected by capital rules → link intact



Finding 3a — Pricing Break Test (2008)

$$R_{i,t}^e = \alpha + \beta_{pre} \cdot LevFac_t + \delta \cdot D_{Post2008} + \beta_{diff} \cdot (LevFac_t \times D_{Post2008}) + \epsilon_{i,t}$$

Panel A: Pricing Break Test (Break Date: 2008, Crisis Onset)			
	Pre-2008	Post-2008	Difference (Post - Pre)
LevFac Beta	0.0097 (1.24)	-0.0454* (-1.86)	-0.0551** (-2.13)

D is a dummy variable that takes the value of one for the post-2008 period and zero otherwise.

- In the Post-2008 sample, the beta does not merely converge to zero but flips to **-0.0454** ($t=-1.86$).
- The significant negative interaction term confirms that the pricing ability of the leverage factor has been structurally destroyed by the financial crisis and the subsequent regulatory environment.

Finding 3b — Volatility Suppression Test (2010)

$$LevFac_t^2 = \gamma_0 + \gamma_{break} \cdot D_{Post2010} + v_t$$

Panel B: Volatility Suppression Test (Break Date: 2010, Regulatory Era)			
	Pre-2010	Post-2010	Difference (Post - Pre)
Constant (Variance)	1.3962*** (11.02)	0.1310*** (12.24)	-1.2652*** (-6.06)

- The variance of the leverage factor innovations has been suppressed by approximately **90%** post-2010.
- A heavily constrained, low-volatility leverage ratio can no longer capture the marginal utility of wealth.

Conclusion & Policy Implications

- **Data Revisions Matter.** Intermediary factor pricing is highly sensitive to ex-post macro data revisions.
- **Theory is Not Dead.** The post-2010 breakdown reflects a regulatory structural break, not a failure of the theory. AEM (2014) perfectly described a world that no longer exists.
- **Policy Implication.** Post-crisis regulations achieved their goal—they successfully suppressed procyclical risk-taking, creating a safer but less informative intermediary sector.

Thank you! I look forward to Fernando's discussion.